

HEATMAP NODE · DESIGN PRECIS

HeatMap Node design precis

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	HMN-PRC-001	Version	0.3
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/heatmap-node		
Source	docs/02-concept.md @ d91cd4d		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3 update: design choices adopted per HMN-DDR-001; numbers checked against HMN-CAL-001; lanyards and FieldNode pole adapter added; parametric model and HMN-DWG-001
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, components, first-order numbers, safety, media, open questions)
0.1	2026-09-25	Amish Chadha	Initial scaffold

HeatMap Node design precis

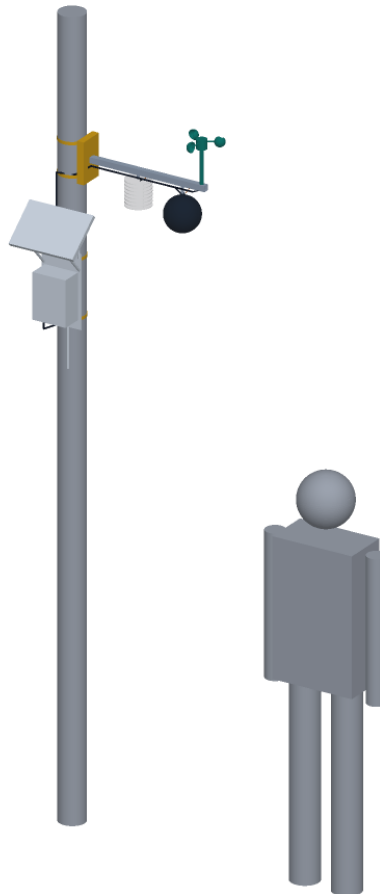
■ Summary

HeatMap Node is a sensor head that clamps to an existing street pole and plugs into the lab's FieldNode core. A horizontal arm holds a naturally ventilated radiation shield with an air temperature and humidity sensor, a standard 150 mm black globe with a thermistor at its center, and a small cup anemometer. From these four readings a server computes mean radiant temperature and an estimated wet bulb globe temperature (WBGT), the heat stress index defined in [ISO 7243:2017](#), using the method of [Liljegren et al. \(2008\)](#). Tens of nodes across a neighborhood give block-by-block heat stress maps, day and night, all season.

The TRL 3 calculations (HMN-CAL-001) confirm the heat stress method and the power, data and wind cases, but six requirements are not met on paper: the passive shield reads about 1 °C high in full sun at 1 m/s (R1), the anemometer does not start below about 0.8 m/s (R4), the FieldNode core overheats at the top of the ambient range (R8), the sensors sit at about 2.7 m rather than pedestrian height (R9), the sensor head costs \$130 against its \$120 budget (R13), and the complete node weighs 4.68 kg against 4 kg (R15). The sensors draw 0.12 mW. The design choices below are adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review (HMN-DDR-001). The parametric model is `cad/src/model.py` and the general arrangement is drawing HMN-DWG-001.

HeatMap Node

CONCEPT, NOT FOR FABRICATION



Grey: existing street pole (not supplied), a 1.75 m person for scale

Figure 1. HeatMap Node on a 114 mm street pole, arm axis at 2.8 m, FieldNode core below it facing the equator, with a 1.75 m person for scale. The pole is not supplied. CONCEPT, NOT FOR FABRICATION.

■ How it works

1. **Air temperature and humidity.** A digital sensor sits inside a stack of eight white plates that block direct and reflected sun while letting wind through. No fan is fitted, to save power and moving parts.
2. **Radiant heat.** A thin copper sphere painted matte black reaches a balance between absorbed sun and long-wave radiation from hot walls and pavement and convective loss to the air. A thermistor at its center reads this globe temperature.
3. **Wind.** A cup anemometer gives the wind speed needed to separate radiant from convective heat at the globe.
4. **Logging and radio.** The FieldNode core samples every 60 s, averages over 15 min, stores the means in flash and sends about 20 bytes by LoRaWAN to a gateway such as TwinKit. Missing packets are resent from flash.
5. **Heat stress and mapping.** A server script computes mean radiant temperature and WBGT and places each node on a block-level map, published as open data. Only environmental values leave the node.

HeatMap Node: data flow from street to map (values from HMN-CAL-001)

CONCEPT, NOT FOR FABRICATION



Figure 2. Data flow from the street to the map. All values are estimates.

HeatMap Node: cutaway

CONCEPT, NOT FOR FABRICATION

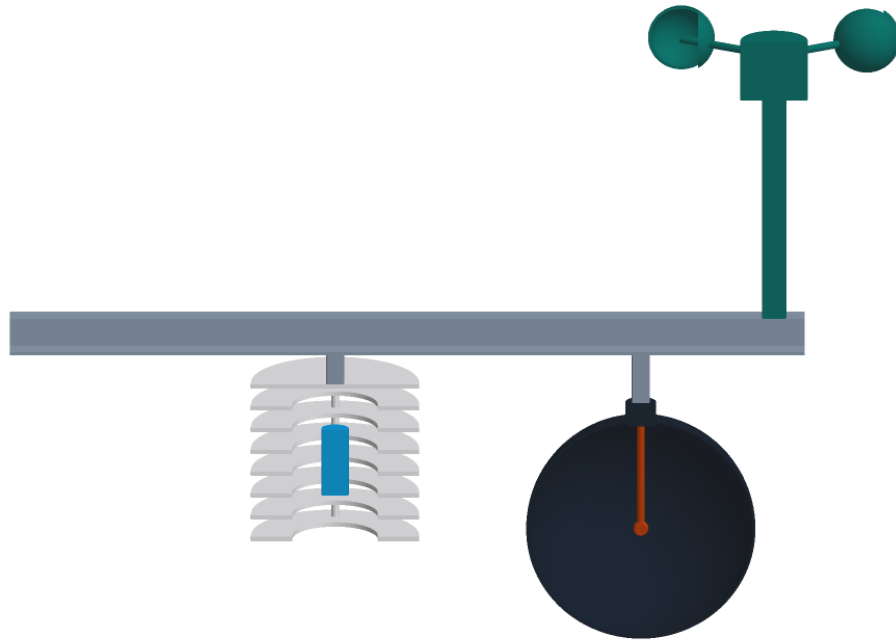


Figure 3. Section through the sensor arm: temperature and humidity sensor inside the radiation shield, and the thermistor at the center of the globe.

■ Main components

Numbers match the exploded view (Figure 4) and `bom/bom.csv`. Line 10 (hardware) has no callout.

#	Component	Choice (TRL 3)	Notes
1	FieldNode core	150 x 90 x 200 mm IP65 enclosure, 6 W panel as sun hood at 40°, 3.2 V 6 Ah LiFePO4 cell, MPPT board, STM32WL-class LoRaWAN module, two M12 ports (FND-DWG-001)	Shared lab component, 2.41 kg, \$126.00 (FND-CAL-001); faces the equator
2	Radiation shield	Eight 110 mm white plates at 15 mm pitch, printed UV-stable ASA	Naturally ventilated
3	Temperature and humidity sensor	SHT45 digital sensor (typical ± 0.1 °C, ± 1.0 %RH) with PTFE membrane cap	I2C over the M12 port
4	Black globe	150 mm copper sphere, about 0.4 mm wall, matte black	Standard globe size, so published globe equations apply
5	Globe probe	10 k NTC bead at the globe center	Calibrated in CalRig
6	Cup anemometer	Three-cup, pulse output, on a short mast at the arm tip	Counted by the FieldNode low-power timer
7	Sensor arm	25 mm square aluminium tube, about 520 mm	Keeps globe and shield about 250 to 450 mm clear of the pole
8	Arm clamp	120° V-saddle 110 x 180 x 30 mm and two 13 mm stainless strap bands for 60 to 200 mm poles	No drilling
9	Sensor harness	Two M12 5-pin leads along the pole and arm	Plug-in at both ends
11	Secondary retention	Two 1.5 mm stainless lanyards, globe boss and shield top to the arm	Added at TRL 3
12	FieldNode pole adapter	Two 120° V-blocks and strap bands in place of FieldNode's 50 mm V-blocks	Added at TRL 3; FieldNode's kit seats only on poles up to about 71 mm

HeatMap Node: exploded view

CONCEPT, NOT FOR FABRICATION

- 1 FieldNode core (FND-DWG-001 envelope)
- 2 Multi-plate radiation shield
- 3 Air temperature and humidity sensor
- 4 Black globe, 150 mm
- 5 Globe temperature probe
- 6 Cup anemometer
- 7 Sensor arm
- 8 Arm clamp and bands
- 9 Sensor harness, M12
- 11 Secondary retention lanyards
- 12 FieldNode pole adapter

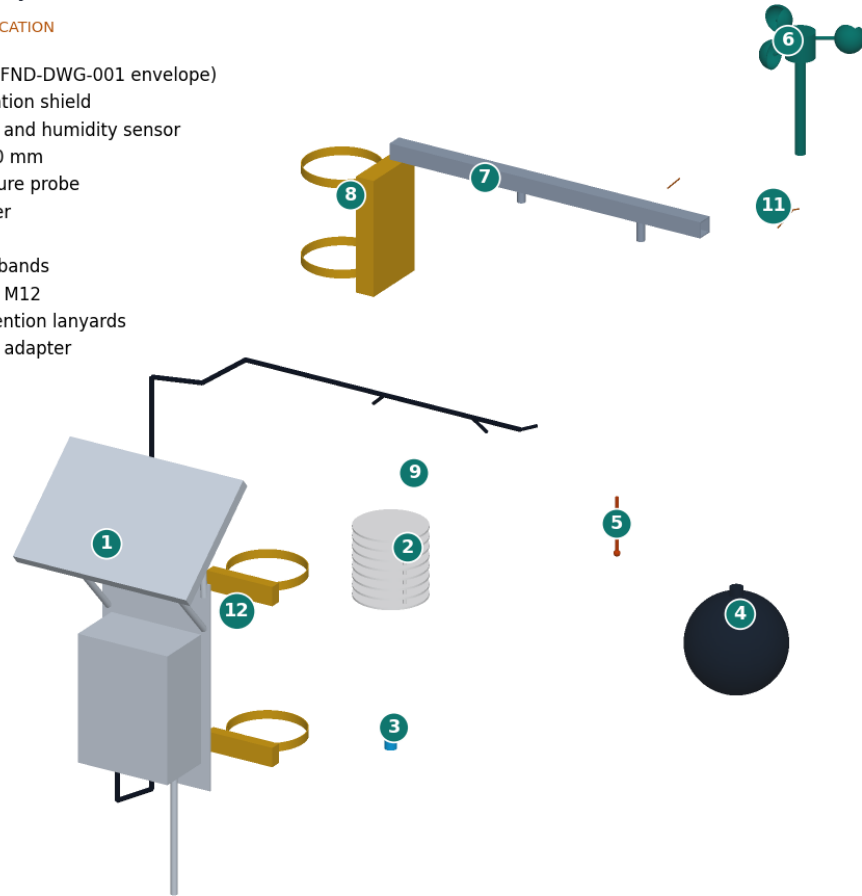


Figure 4. Exploded view with callouts matching the BOM.

The blueprint sheet is in [media/concept-blueprint.pdf](#) and an interactive model in [media/viewer.html](#).

■ First-order numbers

The TRL 2 estimates below have been checked in HMN-CAL-001, and the figures quoted here are from that note. They remain calculations, not measurements.

From measurements to heat stress

For a standard globe under forced convection, mean radiant temperature (MRT) follows from globe temperature T_g , air temperature T_a , wind speed v (m/s), globe diameter D (m) and emissivity ϵ :

$$\text{MRT} = [(T_g + 273)^4 + 1.1 \times 10^8 v^{0.6} / (\epsilon D^{0.4}) \times (T_g - T_a)]^{0.25} - 273$$

This is the forced-convection form of the ISO 7726:1998 method, as implemented in the open [pythermalcomfort](#) library; below about 0.16 m/s in the example natural convection governs and the larger coefficient is used (HMN-CAL-001, section A). Outdoor WBGT is $0.7 T_{nwb} + 0.2 T_g + 0.1 T_a$ (ISO 7243:2017), where the natural wet-bulb temperature T_{nwb} is estimated from the same four readings with the Liljegren method ([open implementation](#)).

Worked example (assumptions: 35 °C air, 40 %RH, 50 °C globe, 1 m/s wind, $D = 0.15$ m, $\varepsilon = 0.95$). MRT is 74.8 °C. The psychrometric wet bulb is 24.2 °C; a wick energy balance puts the natural wet bulb in sun at 26.0 °C, so WBGT is $0.7 \times 26.0 + 0.2 \times 50 + 0.1 \times 35 = 31.7$ °C (HMN-CAL-001, section B). Air temperature alone (35 °C) says nothing about the sun load that the globe captures.

Sensitivity. In the same example, a wind reading of 0.5 m/s instead of 1 m/s lowers MRT to 66.9 °C, and 1.5 m/s raises it to 80.7 °C. A 1 °C air temperature error moves MRT by 1.5 °C. Below the anemometer's start-up speed (about 0.8 m/s) the node reads zero wind, which at a true 0.5 m/s puts MRT 8.2 °C low and WBGT 0.9 °C high. The input errors add up to ± 0.59 °C of WBGT, dominated by the shield's radiation error on air temperature (about 1.0 °C at 1 m/s, R1 not met). A fan-aspirated shield would cut the budget to ± 0.20 °C for about 30 mW; it is proposed in the review note.

Power

The temperature and humidity sensor, sampled once a minute, takes about 11 μ J per reading and the switched thermistor divider about 8 μ J. The anemometer pulses are counted by a low-power timer; with a 100 k Ω pull-up the worst case (reed contact resting closed) is 0.109 mW. The total sensor load is 0.116 mW, 0.12 % of the 100 mW design allowance in FND-CAL-001. HeatMap Node does not need a larger panel or cell.

Radio

At the 15 min default, each node sends 96 uplinks of 20 bytes a day (eleven fields, including the share of each interval below the anemometer's start-up), 23.7 s/day of airtime at SF9 (FND-CAL-001). Seven days of records at FieldNode's 32 B each take 21.5 kB of its 16 MB flash.

Wind load and mass

At a 35 m/s gust (750 Pa) the sensor head carries 44.1 N (globe 6.6 N, shield 10.7 N, anemometer 7.2 N, arm 19.5 N) and the FieldNode core 81 N (FND-CAL-001). The arm root sees 13.7 N·m and 10.5 MPa, a factor of 13.8 on yield; the tip deflects 0.29 mm at 20 m/s. The clamp's band friction resists the twist with a factor of 2.6 at the assumed 1,000 N preload. The node adds about 314 N·m at the pole base, which the pole owner should check.

Mass is 4.68 kg, over the 4 kg of R15: sensor head 2.03 kg (clamp 0.57 kg, arm 0.28 kg, globe 0.28 kg, anemometer 0.30 kg, harness 0.27 kg, shield 0.24 kg and small parts), FieldNode core 2.41 kg (FND-CAL-001; 1.7 kg was assumed at TRL 2) and pole adapter 0.24 kg.

Cost

The sensor head (BOM lines 2 to 12) costs \$130.00 and the full node, with the \$126.00 FieldNode core, \$256.00. Under HMN-DDR-001 D1 the \$120 budget covers the sensor head only, which is \$10 over because of the lanyards and pole adapter added at TRL 3 (R13 not met). See the review note for the options.

■ Key design choices

Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review (HMN-DDR-001).

1. **Build on FieldNode** rather than a separate power and radio design, with a TwinKit gateway or a public LoRaWAN network (D7). Reuses a shared core, costed in its own repo at \$126.00 (D1).
2. **Standard 150 mm globe** rather than a 38 to 40 mm table tennis ball globe. The standard size matches published globe equations and the Liljegren model; the small globe is cheaper and faster to respond but more sensitive to wind and less comparable.

3. **Derived WBGT** rather than a wetted natural wet-bulb sensor, which needs a water reservoir and wick care at every node.
4. **Fit an anemometer** rather than use wind from the nearest official station, because street wind differs strongly from airport wind and MRT is sensitive to it.
5. **Naturally ventilated shield** rather than a fan-aspirated shield. A fan would cut radiation error but adds power, noise and a wear part.
6. **Arm at about 2.8 m** to deter tampering (D2), with a correction to pedestrian height (HMN-CAL-001 estimates 0.2 to 0.8 °C in strong sun) and a later pilot comparing 2.0 and 2.8 m, rather than at 1.5 to 2 m where the data is most representative (R9 not met).
7. **Open data, environmental channels only** (R12, D8). Who publishes and hosts the data remains open.

■ Safety

Safety: Mounting on street poles is work at height next to traffic. Install only with the pole owner's written permission, by trained crews, with a stable ladder or lift, fall protection and traffic management as local rules require, and keep clear of overhead power lines and any live parts of lighting poles. Check that the pole can carry the added wind load.

Safety: The FieldNode core contains a LiFePO₄ cell of about 19 Wh. Fuse it at the holder, charge only within the maker's temperature limits, and do not install a swollen, damaged or wet cell.

Safety: The globe and the shield hang from the arm on fixings backed by stainless lanyards, so a failed fixing does not drop a part onto the street. Check the fixings and lanyards at every visit.

Safety: The black globe can reach about 67 °C at 50 °C air in full sun (HMN-CAL-001), and the arm runs hot too. Let them cool or wear gloves before handling. Cut edges on the aluminium arm and copper sphere must be deburred, and anemometer cups spin; stop them before working near them.

Safety: HeatMap Node data is for planning and research. It is not an official heat warning, and it must not be used to decide whether a particular person is safe to work or exercise without the responsible agency's guidance.

■ Open questions

- [] Budget: the sensor head is \$10 over its \$120 budget after the TRL 3 additions (see the review note).
- [] Mounting height: what do pole owners allow? The calculated difference (0.2 to 0.8 °C) needs a field comparison.
- [] Shield: HMN-CAL-001 puts the passive shield's error at about 1 °C at 1 m/s; is the 30 mW fan option worth it?
- [] Which anemometer meets R4 at the low end at this price?
- [] Globe response: HMN-CAL-001 estimates about 12 min to 90 %, faster than the TRL 2 estimate; a step test is needed.
- [] How often must nodes be recalibrated in CalRig, and how are dust and fading of the black paint handled?
- [] Data model and hosting: TwinKit, CityTwin or a public platform, and who publishes it?
- [] Night-time value: are 15 min means enough to capture nights that do not cool, when the body cannot recover from daytime heat?